# Final Year Trip

Time limit per test: 2 seconds
Memory limit per test: 256 mb

Owais Idrees is planning on finalising the Final Year Trip for the 23 Batch of FAST Islamabad. It is the last trip of their degree, the one everybody has been talking about since the second semester, and Owais has decided that it will be remembered for the right reasons. He spent three weeks going through every travel page, every vlog and every group chat recommendation he could find, and finally made a list of all the locations he wants the trip to go to — the viewpoints, the waterfalls, the lakes, the food streets, and the one hotel everybody keeps sending pictures of.

But the issue is that the trip will go via coasters provided by Naveed Chacha under **Uzair Transport Service**. Naveed Chacha is really touchy about his coasters, even though they are not up to the quality they should be. He has been driving students up north for eighteen years, and in those eighteen years he has developed exactly one opinion that he repeats at every single meeting:

> **Naveed Chacha:** *"My coasters will not go through any roads that will cause damage to the coaster shocks or anywhere in the engine. You people sit at the back with your speakers and your TikToks, I am the one who pays the mechanic."*

So, to make sure that no such route is taken, Naveed Chacha sits down with Owais one evening at the transport office, takes out a register that is older than most of the students, and writes down a list of all the roads that are safe to travel and are approved by him. These roads are good and smooth for the coaster to travel in, and Naveed Chacha guarantees that the coaster can go in either direction on any of them.

Owais Idrees was satisfied with the proposition and reached for the pen. That is when Naveed Chacha turned the register around and pointed at the column on the right, the one with the rates.

> **Naveed Chacha:** *"One more thing. I do not charge you by the day, I charge you by the road. Every road the coaster drives, that is diesel, that is the shocks, that is my mechanic. And if you drive the same road again on the way back, that is a second road on the bill. I am not running a charity."*

Owais Idrees was satisfied with this too, because he could simply plan a short route. But the problem came when he went to **Hasan Mujtaba** for approval. Hasan Mujtaba read the plan twice, put it down, and added one more special condition.

> **Hasan Mujtaba:** *"These students are never happy. Every time they complain about the travel and everything. Last year you took them to four places out of nine and the feedback forms were a disaster. So this time, every single location on that list gets visited. Not most of them. All of them."*

> **Owais Idrees:** *"Sir, some of those places are only reachable by going back through Murree. I will end up driving the same road two or three times."*

> **Hasan Mujtaba:** *"Then drive it two or three times, I do not care how the coaster gets there. Drive a road as often as you like, pass through a town as often as you like. But the coaster leaves from the FAST gate, it touches every location on the list, and it comes back to the FAST gate. And since Naveed Chacha is counting roads, you had better find the cheapest way to do it."*

Now this complicates things. The coaster may drive any approved road as many times as it needs to, and it may pass through the same location again and again, but every road it drives is another line on Naveed Chacha's bill. Owais Idrees was really confused on what to do now, so he contacted the one person who could fix the problem. The one person with the right *jugaar* for the problem, **Jawad Ahmad Khan**.

Jawad saw the problem and immediately started working on the solution. He first tried entering each of the locations into Google Maps to find a good route, but it did not take the roads as input at all — it kept suggesting shortcuts on roads that Naveed Chacha had never approved, and it had no idea that the bill counts every single road driven. He spent 5 hours dragging pins around on a laptop screen and came up with a route that visited all the locations but drove eleven roads where seven would have done.

He leaned back, looked at the ceiling, and thought to himself:

> **Jawad Ahmad Khan:** *"I spent 4 years completing my CS degree. If I cannot solve this problem then all that goes to waste."*

He takes out his laptop, closes every Maps tab, and solves it via code in only 15 minutes.

Your task is to do what Jawad did. Given the list of locations and the list of roads approved by Naveed Chacha, find a route that

- starts at the location named `FAST-gate-1` and finishes there,
- visits every location at least once, and
- drives as few roads as possible.

A road may be driven any number of times and each drive counts separately towards the total. A location may be passed through any number of times. If some location simply cannot be reached from the FAST gate using the approved roads, the trip is off.

## Input:

The first input line has two integers $\boldsymbol{n}$ and $\boldsymbol{m}$: the number of locations and the number of approved roads. The locations are numbered 1, 2, ..., $n$.

The next line contains $n$ different location names separated by spaces, given in order from location 1 to location $n$. Each name is a non-empty string of at most 20 characters and contains no spaces. Exactly one of them is `FAST-gate-1`, and it is not necessarily the first one.

After that, there are $m$ lines describing the roads. Each line has two integers $\boldsymbol{a}$ and $\boldsymbol{b}$: there is an approved road between location $a$ and location $b$. All roads are two-way roads. Every road is between two different locations, and there is at most one approved road between any pair of locations.

## Output:

If every location can be reached from `FAST-gate-1`, print two lines.

On the first line print a single integer $K$: the smallest number of roads the coaster can drive. On the second line print the route as $K + 1$ location **names** separated by spaces, in the order they are visited. The first and the last name must both be `FAST-gate-1`, every pair of consecutive names must have an approved road between them, and every location must appear at least once.

If more than one route drives $K$ roads, print any of them.

If some location cannot be reached from `FAST-gate-1`, print a single line containing "IMPOSSIBLE".

## Constraints:

- $2 \leq n \leq 16$
- $1 \leq m \leq \frac{n(n-1)}{2}$
- $1 \leq a, b \leq n$ and $a \neq b$
- Each location name has length at most 20 and contains no whitespace.

## Example:

### Example 1

Input

```
6 8
FAST-gate-1 murree naran kaghan shogran abottabad
1 2
1 3
2 3
2 4
2 6
3 5
3 6
4 5
```

Output

```
7
FAST-gate-1 murree abottabad naran shogran kaghan murree FAST-gate-1
```

**Explanation:**

The coaster drives seven roads:

| # | Road driven | From → To |
|---|---|---|
| 1 | 1 – 2 | FAST-gate-1 → murree |
| 2 | 2 – 6 | murree → abottabad |
| 3 | 6 – 3 | abottabad → naran |
| 4 | 3 – 5 | naran → shogran |
| 5 | 5 – 4 | shogran → kaghan |
| 6 | 4 – 2 | kaghan → murree |
| 7 | 2 – 1 | murree → FAST-gate-1 |

All six locations are visited. Murree is passed through twice and the road between the FAST gate and Murree is driven twice, once in each direction — both are allowed, and the second drive is counted in the total of seven. No route visits all six locations in six roads, so seven is the minimum.

## Example 2

Input

```
5 4
FAST-gate-1 murree naran nathiagali ayubia
1 2
2 3
3 1
4 5
```

Output

```
IMPOSSIBLE
```

**Explanation:**

Nathiagali and Ayubia are joined to each other but to nothing else, so no approved road leads there from the FAST gate. Hasan Mujtaba will not accept a route that skips them, and Naveed Chacha will not add a road, so the trip is off.

## Example 3

Input

```
5 5
murree naran kaghan FAST-gate-1 shogran
1 2
2 3
3 4
4 5
5 1
```

Output

```
5
FAST-gate-1 shogran murree naran kaghan FAST-gate-1
```

**Explanation:**

Here the FAST gate is location 4, not location 1. The five roads form a single loop, so the coaster drives all the way around it once, visiting every location exactly once and returning to the gate after five roads. Driving the loop in the other direction is also a correct answer.

## Notes:

- The route must both start and end at `FAST-gate-1`.
- A road may be driven more than once, and each drive counts towards $K$.
- A location may be visited more than once. Only the total number of roads matters.
- Not every approved road has to be used. Roads that lead nowhere useful can be left alone.
- If several routes drive the minimum number of roads, any one of them is accepted.